

Professional Self Assessment

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Completing the Computer Science program at Southern New Hampshire University and developing my ePortfolio have been defining experiences in shaping both my technical abilities and my professional identity. Throughout the program, I have grown from a student learning foundational concepts into a developer capable of designing, engineering, and improving real world software systems. My capstone project, which focused on enhancing and stabilizing a mobile Weight Tracking App, served as the culmination of this growth and allowed me to demonstrate my strengths in software engineering, debugging, database design, and user focused development.

How the Program and ePortfolio Shaped My Strengths and Professional Goals

Across my coursework, I learned to approach problems systematically, communicate effectively, and build solutions that are both technically sound and centered on the needs of the user. Courses in algorithms, full stack development, databases, and software architecture gave me the foundation to understand how systems behave and how to improve them. The ePortfolio process allowed me to showcase these strengths in a cohesive and professional format that highlights not only what I built, but how I think as a developer.

My capstone project reinforced my long term goal of becoming a mobile and full stack developer. Enhancing the Weight Tracking App required me to apply the full range of my skills, including analyzing existing code, identifying root causes of crashes, improving user interface flows, restructuring database logic, and adding new features such as the calorie goal calculator. This experience strengthened my confidence in tackling unfamiliar codebases and delivering meaningful improvements, which directly supports my employability in the software development field.

Professional Skills Developed Throughout the Program

Collaborating in a Team Environment Although the capstone is an individual project, collaboration was central throughout the program. Group assignments in earlier courses taught me how to divide responsibilities, communicate progress, and integrate code across multiple contributors. These experiences helped me develop habits that carried into my solo work, including writing clear documentation, organizing code logically, and thinking about how others would read, maintain, or extend my work.

Communicating with Stakeholders Many assignments required me to translate technical decisions into language that is understandable to non technical audiences. Whether writing design documents, explaining architectural choices, or presenting project updates, I learned to communicate clearly, concisely, and professionally. This skill was essential in the capstone, where I documented the app issues, explained enhancement decisions, and justified the inclusion of each artifact in my portfolio.

Data Structures and Algorithms The program strengthened my understanding of algorithmic thinking and efficient data handling. While the Weight Tracking App did not require advanced algorithms, the logic behind the calorie goal calculator, data validation routines, and user interface driven workflows all relied on algorithmic reasoning. My coursework in algorithms prepared me to write code that is not only functional but optimized and maintainable.

Software Engineering and Databases Software engineering principles guided every enhancement I made to the Weight Tracking App. I applied debugging strategies, improved activity lifecycle management, corrected user interface bindings, and refined navigation flows. Database coursework directly supported my ability to fix SQL schema mismatches, improve

create read update delete operations, and stabilize the app data layer. These experiences demonstrate my readiness to work on production level systems.

Security Security principles were woven throughout the program, from secure coding practices to understanding vulnerabilities and data protection. In the capstone, I applied these principles by ensuring that no sensitive information was stored improperly, avoiding exposure of credentials, and maintaining safe handling of user data. My coursework prepared me to think proactively about security, even in small applications.

How My Artifacts Fit Together in the Portfolio

The artifacts in my ePortfolio collectively demonstrate the breadth of my computer science abilities. The enhanced Weight Tracking App showcases my strengths in mobile development, debugging, user interface design, and database engineering. Earlier projects, including full stack applications, algorithmic assignments, and database designs, highlight my versatility across multiple domains of computer science.

Together, these artifacts tell a cohesive story.

- I can design software.
- I can build software.
- I can debug and improve existing systems.
- I can communicate my decisions clearly.
- I can apply computer science principles to real problems.

This portfolio reflects not only what I have learned, but who I have become as a developer. I am capable, adaptable, and ready to contribute meaningfully in a professional environment.